

ChronoCare

From scattered data to clinical clarity — in seconds.

14 MCP Tools	8 LLM Calls / Pipeline	14–16s Warm Latency	\$0.09 / Analysis
Python 3.11	OpenAI + Groq	FHIR R4	MCP + A2A

1. Executive Summary

ChronoCare is a clinical reasoning engine that fills the two missing tenses of patient care. Electronic health records answer one question well — what is happening right now. They systematically miss two more: what has been happening, and what is coming. ChronoCare answers all four critical clinical questions in a single conversation, in under 15 seconds, using a 13-step AI reasoning pipeline built on open healthcare standards.

The system combines 14 Model Context Protocol (MCP) tools, 8 chained LLM calls routed across OpenAI and Groq, a FHIR R4 data layer, and an A2A agent on the Prompt Opinion platform. It is fully deployed in production with a React web frontend, automated CI/CD, and four synthetic demo patients covering distinct clinical archetypes.

2. Problem Statement

Modern EHRs are snapshot systems. They show a clinician what is true at this moment. What they cannot show — structurally, not through any lack of data — is the trajectory. Consider:

Gap	Example	Consequence
Trajectory blindness	Creatinine drifts 0.9 → 1.3 mg/dL over 36 months. Each time a clinician sees the snapshot, they think it's new.	Three different doctors see the same snapshot. None see the trend.
Cascade invisibility	Hypertension (2019) → Stage 2 CKD (2022). No system flagging the developing interval.	End-stage disease develops in silence.
Weak signal blindness	BP 138/88 (normal) + Creatinine 1.3 (normal) + fatigue (patient history) = missed signals.	Pat 2 misses the signal to miss all signals together.

Impact: Diagnostic errors are the 3rd leading cause of death in the United States (40,000–80,000 deaths/year). Handoff failures cause 80% of serious medical errors. These are information architecture failures, not data availability failures.

3. Solution: The Four-Layer Clinical Brief

#	Question	Module	Output
01	What happened?	Time Traveler	Chronological narrative, 5 turning points
02	What is going wrong?	Silent Deterioration Detector	Risk score, weak signal analysis
03	Why did this happen?	Root Cause Analyzer	Causal hypothesis with confidence
04	What should we do?	Recommendation Engine	3–5 guideline-cited actions

4. System Architecture

Four replaceable layers held together by open standards:

Layer	Technology	Role
Frontend	React 18 + TypeScript + Vite · Tailwind CSS · Framer Motion · GitHub Pages	Live demo UI, animated pipeline trace, admin panel
Orchestration	ChronoCore A2A Agent · Prompt Opinion · GPT-4.1	Receives natural-language requests, delegates to MCP
Reasoning	Python 3.11 · Starlette + Uvicorn · MCP SDK · Railway	14 MCP tools, fan-out/fan-in parallel pipeline, FHIR client
Data & Capacity	HAPI FHIR R4 · OpenAI (GPT-4o, mini, TTS-1) · Groq (Llama-3.3-70b)	Patient data, LLM reasoning + voice synthesis

5. The 13-Step Reasoning Pipeline

The pipeline uses a fan-out / fan-in topology. After a sequential head (FHIR fetch + chronological ordering), five independent branches execute concurrently via `asyncio.gather`. This dropped wall-clock latency from 25–35s sequential to 14–16s parallel.

#	Tool	Engine	Phase
01	get_full_patient_history	FHIR REST	Reconstruct
02	order_events_chronologically	Pure Python	Reconstruct
03	identify_clinical_turning_points	Llama-3.3-70b	Reconstruct
04	generate_patient_narrative	GPT-4o	Reconstruct
05	get_recent_signals	Pure Python	Detect
06	analyze_weak_patterns	GPT-4o	Detect
07	generate_early_warning_report	Llama-3.3-70b	Detect
08	correlate_events	GPT-4o-mini	Explain
09	generate_causal_hypothesis	GPT-4o	Explain
10	map_comorbidities	Llama-3.3-70b	Recommend
11	match_clinical_guidelines	GPT-4o-mini	Recommend
12	generate_recommendations	GPT-4o	Recommend
13	generate_unified_brief	Pure Python	Synthesize

14	text_to_speech_brief (opt.)	OpenAI TTS-1	Voice
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6. Multi-Model LLM Routing

8 LLM calls per pipeline routed across two providers based on task characteristics. GPT-4o handles the four tasks where output quality directly affects clinical credibility. Llama-3.3-70b handles structured extraction where speed matters more than prose. Multi-provider routing also provides partial-failure resilience.

Task	Model	Provider	Rationale
Clinical narrative	GPT-4o	OpenAI	Best prose quality
Silent deterioration analysis	GPT-4o	OpenAI	Critical reasoning depth
Causal hypothesis	GPT-4o	OpenAI	Nuanced multi-step logic
Recommendations	GPT-4o	OpenAI	Consequential output
Turning points + Warning	Llama-3.3-70b	Groq	Fast structured JSON
Comorbidity mapping	Llama-3.3-70b	Groq	Speed over prose quality
Event correlation	GPT-4o-mini	OpenAI	Accuracy at lower cost
Guideline matching	GPT-4o-mini	OpenAI	Factual retrieval
Voice synthesis	TTS-1	OpenAI	Natural clinical prosody

Cost: ~\$0.09 per full analysis. Approximately 10 cents per patient brief.

7. Synthetic Patient Data

All patient data is fully synthetic. No PHI anywhere in the system. Four demo patients are hosted on the HAPI public FHIR R4 sandbox:

Patient	Age/Sex	Clinical Archetype	Risk
John Doe	62M	Hypertension (2019) → Stage 2 CKD (2022). Original silent cardiometabolic syndrome.	HIGH
Maria Rodriguez	58F	T2DM since 2020. SGLT2 added 2023. UACR rising. Early diabetic nephropathy.	MED
Robert Chen	71M	CHF (EF 35%) + AFib + COPD. Cardiorenal syndrome. BNP rising, eGFR declining.	HIGH
Sarah Williams	45F	Prediabetes + borderline HTN. Lifestyle success. Control case — tests not flagging.	LOW

FHIR resources used: Patient, Condition, Observation, MedicationRequest, Encounter, DocumentReference, DiagnosticReport. Code systems: LOINC (37 codes), ICD-10 (32 codes), RxNorm (30 codes).

8. Verified Production Metrics

Not aspirational. Observed in production on the live Railway deployment:

Metric	Value	How Verified
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End-to-end latency (warm)	14–16 seconds	Repeated runs on John Doe via /api/demo/analyze
End-to-end latency (cold)	~30 seconds	First request after Railway idle/redeploy
MCP tools enumerable	14 / 14	tools/list JSON-RPC against /mcp/
Backend tests passing	42 / 42	pytest tests/ -v
FHIR resource types / request	7 in parallel	Logs show 7 GETs to HAPI per fetch
LLM calls per pipeline	8 distinct	Logs show 8 llm_call events, mixed models
Frontend bundle size	~110 KB gzipped	vite build artifact
Demo patients on HAPI	4 / 4	All retrievable via GET /Patient/<id>
Pipeline success rate	100%	All patients, all runs, trace ok:true
Cost per full analysis	~\$0.09 USD	Token counts × per-model pricing

9. Full Technology Stack

Category	Technologies
AI & LLMs	OpenAI GPT-4o · GPT-4o-mini · TTS-1 · Groq Llama-3.3-70b-versatile
Backend	Python 3.11 · Starlette · Uvicorn · MCP SDK · httpx · structlog · slowapi
Frontend	React 18 · TypeScript · Vite 5 · Tailwind CSS 3 · Framer Motion 11 · React Router 6
Data & Standards	HAPI FHIR R4 · LOINC · ICD-10 · RxNorm · MCP Protocol · A2A · SHARP Context
Infrastructure	Railway · GitHub Pages · GitHub Actions CI/CD · Docker · Docker Compose
Testing	pytest · pytest-asyncio · 42 unit + integration tests · Live smoke checks in CI
Logging & Auth	structlog JSON · SHA-256 hashed patient IDs · API key header auth

10. Live URLs

Surface	URL
Frontend (GitHub Pages)	https://sriksven.github.io/chronocare/
Backend (Railway)	https://attractive-ambition-production-5fd7.up.railway.app
MCP Endpoint	/mcp/ (Streamable HTTP, trailing slash required)
Public Analyze API	POST /api/demo/analyze
Public Voice API	POST /api/demo/voice
Health Check	GET /health
FHIR Data Store	https://hapi.fhir.org/baseR4 (HAPI public R4 sandbox)
Source Repository	https://github.com/sriksven/chronocare

11. What's Next for ChronoCare

Priority	Initiative	Description
1	Live EHR Integration	Test against Epic + Cerner sandboxes. One env var change.
2	Specialty Modules	Oncology, psychiatry, pediatrics with domain-specific reasoning.
3	Streaming Output	Deliver each brief section as it completes — narrative first.
4	Longitudinal Monitoring	Daily scheduled runs, diff against prior brief, surface changes.
5	Federated Deployment	Self-host inside hospital infra. No data leaves the network.
6	Fine-Tuned Model	Clinical reasoning model trained on verified outputs.

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github.com/sriksven/chronocare · sriksven.github.io/chronocare